

*SEE NOTES		TIE ROD SCHEDULE											
TEST PRESSURE		170 kPa TIE RODS		340 kPa TIE RODS		690 kPa TIE RODS		1035 kPa TIE RODS		1550 kPa TIE RODS		2585 kPa TIE RODS	
PIPE DIA (mm)	MIN PIPE WALL THKS (mm)*	DIA (mm)	NO. REQD	DIA (mm)	NO. REQD	DIA (mm)	NO. REQD	DIA (mm)	NO. REQD	DIA (mm)	NO. REQD	DIA (mm)	NO. REQD
150	4.8	—	—	—	—	16	2	16	2	16	2	162	2
200	4.8	—	—	—	—	16	2	16	2	16	2	202	2
250	4.8	—	—	—	—	16	2	16	2	20	2	222	2
300	4.8	16	2	16	2	16	2	16	2	20	2	222	4
350	4.8	16	2	16	2	20	2	20	2	20	4	252	4
400	4.8	16	2	16	2	20	2	22	2	22	4	252	4
450	6.4	16	2	16	2	22	2	25	2	25	4	322	4
500	6.4	16	2	20	2	22	2	22	4	25	4	324	4
550	6.4	16	2	20	2	20	4	22	4	25	4	384	4
600	6.4	16	2	20	2	22	4	25	4	25	6	384	6
750	6.4	16	4	20	4	22	6	25	6	25	8	386	8
900	6.4	20	4	22	4	25	6	25	8	32	8	388	10
1050	6.4	20	4	25	4	25	8	32	8	38	8	448	10
1200	7.9	22	4	22	8	25	10	32	10	38	10	44	12
1350	7.9	20	6	22	8	25	12	32	12	38	12	44	14
1500	8.7	22	6	25	8	32	10	32	14	38	14	44	16
1800	9.5	22	8	25	10	32	12	38	14	44	14	51	18
1950	10.3	22	8	25	12	32	14	38	16	44	16	51	20
2100	11.1	22	10	32	10	38	14	44	14	44	18	51	22
2400	11.1	25	10	32	12	38	16	44	16	51	16	—	—
2700	12.7	25	12	32	12	44	12	51	14	51	22	—	—
3000	15.9	32	8	38	10	44	16	51	18	—	—	—	—
3300	18.3	32	6	44	10	44	18	51	22	—	—	—	—

NOTES:

1. ALL DIMENSIONS IN MILLIMETERS, UNLESS OTHERWISE NOTED.
2. TIE RODS SHALL CONFORM TO ASTM A193 GRADE B7.
3. NUTS SHALL CONFORM TO ASTM A194 GRADE 2H.
4. TIE ROD NUTS SHALL BE TIGHTENED GRADUALLY AND EQUALLY IN STAGES TO PREVENT UNEVEN ALIGNMENT AND TO ALLOW EQUAL STRESS ON ALL TIE RODS UNDER PRESSURE. TIGHTEN UNTIL SNUG. THREADS SHALL PROTRUDE FROM NUTS. PEEN THREADS AFTER TIGHTENING NUTS.
5. CATHODIC PROTECTION FOR FLEXIBLE COUPLINGS REQUIRED WHERE NOTED ON DWGS OR IN THE SPECIFICATIONS.
6. CONTRACTOR SHALL USE DATA FOR ONLY THOSE PIPE SIZES AND TEST PRESSURES SPECIFIED IN THIS CONTRACT.
7. TIE RODS SHALL NOT BE ATTACHED TO A PIPE WHEN THE WALL THICKNESS IS LESS THAN THE MINIMUM SHOWN ON THE TIE ROD SCHEDULE.
8. THE MINIMUM PIPE WALL THICKNESSES SHOWN ARE TO ENSURE PROPER PERFORMANCE OF THE THRUST TIE LUG. PIPE WALL THICKNESSES GREATER THAN SHOWN IN THE TABLE MAY BE REQUIRED AND MAY BE SHOWN ELSEWHERE OR SPECIFIED ELSEWHERE TO RESIST INTERNAL PRESSURES.

SPECIAL THRUST TIE FOR STEEL PIPE - TIE ROD SCHEDULE

NTS

STANDARD DETAIL

SHT 2 OF 3

15540B

JACOBS